## §2.5 矩阵分块法

### 一、定义：

将矩阵A用若干条纵线和横线分成许多个小矩阵，每一个小矩阵称为A的子块，以子块为元素的形式上的矩阵称为分块矩阵.

eg: 3×4矩阵：

$$A=\begin{pmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \end{pmatrix}$$

3种分法：

(i) $\left(\begin{array}{cc|cc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$ (ii) $\left(\begin{array}{c|cc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$ (iii) $\left(\begin{array}{c|c|c|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$

$\downarrow$

$$A=\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

### 二、运算

#### 1. 加法

$$A=\left(\begin{array}{cc|cc} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ \hline A_{31} & A_{32} & A_{33} & A_{34} \end{array}\right) \qquad B=\left(\begin{array}{cc|cc} B_{11} & B_{12} & B_{13} & B_{14} \\ B_{21} & B_{22} & B_{23} & B_{24} \\ \hline B_{31} & B_{32} & B_{33} & B_{34} \end{array}\right)$$

$A_{11}$ $A_{12}$ $A_{21}$ $A_{22}$ $B_{11}$ $B_{12}$ $B_{21}$ $B_{22}$

$$A+B=\left(\begin{array}{c|c} A_{11}+B_{11} & A_{12}+B_{12} \\ \hline A_{21}+B_{21} & A_{22}+B_{22} \end{array}\right)$$

两个矩阵：同型矩阵，并且采用相同的分块法才能相加.

2. 数乘.

$$A=\begin{array}{c} \begin{array}{cc} A_{11} & A_{12} \end{array} \\ \left(\begin{array}{cc|cc} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ \hline A_{31} & A_{32} & A_{33} & A_{34} \end{array}\right) \\ \begin{array}{cc} A_{21} & A_{22} \end{array} \end{array} \qquad \lambda A=\begin{pmatrix} \lambda A_{11} & \lambda A_{12} \\ \lambda A_{21} & \lambda A_{22} \end{pmatrix}$$

3. 乘法

设A为 $m\times l$ 矩阵, B为 $l\times n$ 矩阵 把A,B分块如下.

$$A=\begin{array}{c} m_1 \\ m_2 \\ \vdots \\ m_s \end{array}\begin{pmatrix} A_{11} & A_{12} & \cdots & A_{st} \\ \vdots & \vdots & & \vdots \\ A_{s1} & A_{s2} & \cdots & A_{st} \end{pmatrix} \qquad B=\begin{array}{c} l_1 \\ l_2 \\ \vdots \\ l_t \end{array}\begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1r} \\ \vdots & \vdots & & \vdots \\ B_{t1} & B_{t2} & \cdots & B_{tr} \end{pmatrix}$$

$$\begin{array}{cccc} l_1 & l_2 & \cdots & l_t \end{array} \qquad \begin{array}{ccc} n_1 & n_2 & \cdots n_r \end{array}$$

$$C=AB=\begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1r} \\ \vdots & \vdots & & \vdots \\ C_{s1} & C_{s2} & \cdots & C_{sr} \end{pmatrix} \qquad C_{ij}=\sum_{k=1}^{t} A_{ik}B_{kj}$$

$(i=1,\cdots,s)$
$(j=1,\cdots,r)$

① 其中 $A_{i1},A_{i2},\cdots A_{it}$ 的列数分别等于 $B_{1j},B_{2j},\cdots B_{tj}$ 的行数.

② 分完之后 A 与B 的分块 整体上 A子块的列数等于B子块的行数相等

4. 转置

$$A=\begin{pmatrix} A_{11} & \cdots & A_{1r} \\ \vdots & \ddots & \vdots \\ A_{s1} & \cdots & A_{sr} \end{pmatrix} \qquad A^T=\begin{pmatrix} A_{11} & \cdots & A_{s1} \\ \vdots & \ddots & \vdots \\ A_{1r} & \cdots & A_{sr} \end{pmatrix}$$

分块矩阵不仅形式上进行转置，而且每个子块也进行转置.

eg:

$$A=\left(\begin{array}{c|c|c|c} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \end{array}\right)=(a_1, a_2, a_3, a_4)$$

$$A^T=\left(\begin{array}{ccc} A_{11} & A_{21} & A_{31} \\ \hline A_{12} & A_{22} & A_{32} \\ \hline A_{13} & A_{23} & A_{33} \\ \hline A_{14} & A_{24} & A_{34} \end{array}\right)=\begin{pmatrix} a_1^T \\ a_2^T \\ a_3^T \\ a_4^T \end{pmatrix}$$

5. 分块对角矩阵

定义：A是n阶矩阵：

①：A的分块矩阵只有在对角线上有非零子块.

②：其余子块都为零矩阵.

③：对角线上的子块都是方阵

那么称A为分块对角矩阵.

性质：$|A|=|A_1||A_2|\cdots|A_s|$

若$|A_i|\neq 0\ (i=1,2,\cdots,s)$，则$|A|\neq 0$ 并有.

$$A^{-1}=\begin{pmatrix} A_1^{-1} & & & 0 \\ & A_2^{-1} & & \\ & & \ddots & \\ 0 & & & A_s^{-1} \end{pmatrix}$$

三. 特殊分块.

换行分块以及换列分块.

矩阵$A_{m\times n}$ 若干等于行的作

$\alpha_i^T=(a_{i1}, a_{i2}, \cdots a_{in})$ $|a_{ij}|$ m个行向量

$$\beta_j = \begin{pmatrix} a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$$ 为列…

则

$$A = \left(\begin{array}{c|c|c|c} a_{11} & a_{12} & \cdots & a_{1n} \\ \hline a_{21} & a_{22} & \cdots & a_{2n} \\ \hline \vdots & \vdots & & \vdots \\ \hline a_{m1} & a_{m2} & \cdots & a_{mn} \end{array}\right) = \begin{pmatrix} \alpha_1^T \\ \alpha_2^T \\ \vdots \\ \alpha_m^T \end{pmatrix} = (\beta_1, \beta_2, \cdots, \beta_n)$$

A按行分块

A按列分块.

按行分块按列分块与矩阵乘积的关系.

于是设 $A$ 为 $m\times s$ 矩阵，$B$ 为 $s\times n$ 矩阵，

$$C=(c_{ij})_{m\times n}=AB=\begin{pmatrix}\alpha_1^T\\ \alpha_2^T\\ \vdots\\ \alpha_m^T\end{pmatrix}(\beta_1,\beta_2,\cdots,\beta_n)=\begin{pmatrix}\alpha_1^T\beta_1 & \alpha_1^T\beta_2 & \cdots & \alpha_1^T\beta_n\\ \alpha_2^T\beta_1 & \alpha_2^T\beta_2 & \cdots & \alpha_2^T\beta_n\\ \vdots & \vdots & & \vdots\\ \alpha_m^T\beta_1 & \alpha_m^T\beta_2 & \cdots & \alpha_m^T\beta_n\end{pmatrix}$$

$$c_{ij}=\alpha_i^T\beta_j=(a_{i1},a_{i2},\cdots,a_{is})\begin{pmatrix}b_{1j}\\ b_{2j}\\ \vdots\\ b_{sj}\end{pmatrix}=\sum_{k=1}^{s}a_{ik}b_{kj}.$$

矩阵 $A=O$ 的充分必要条件是 $A^TA=O$

零矩阵